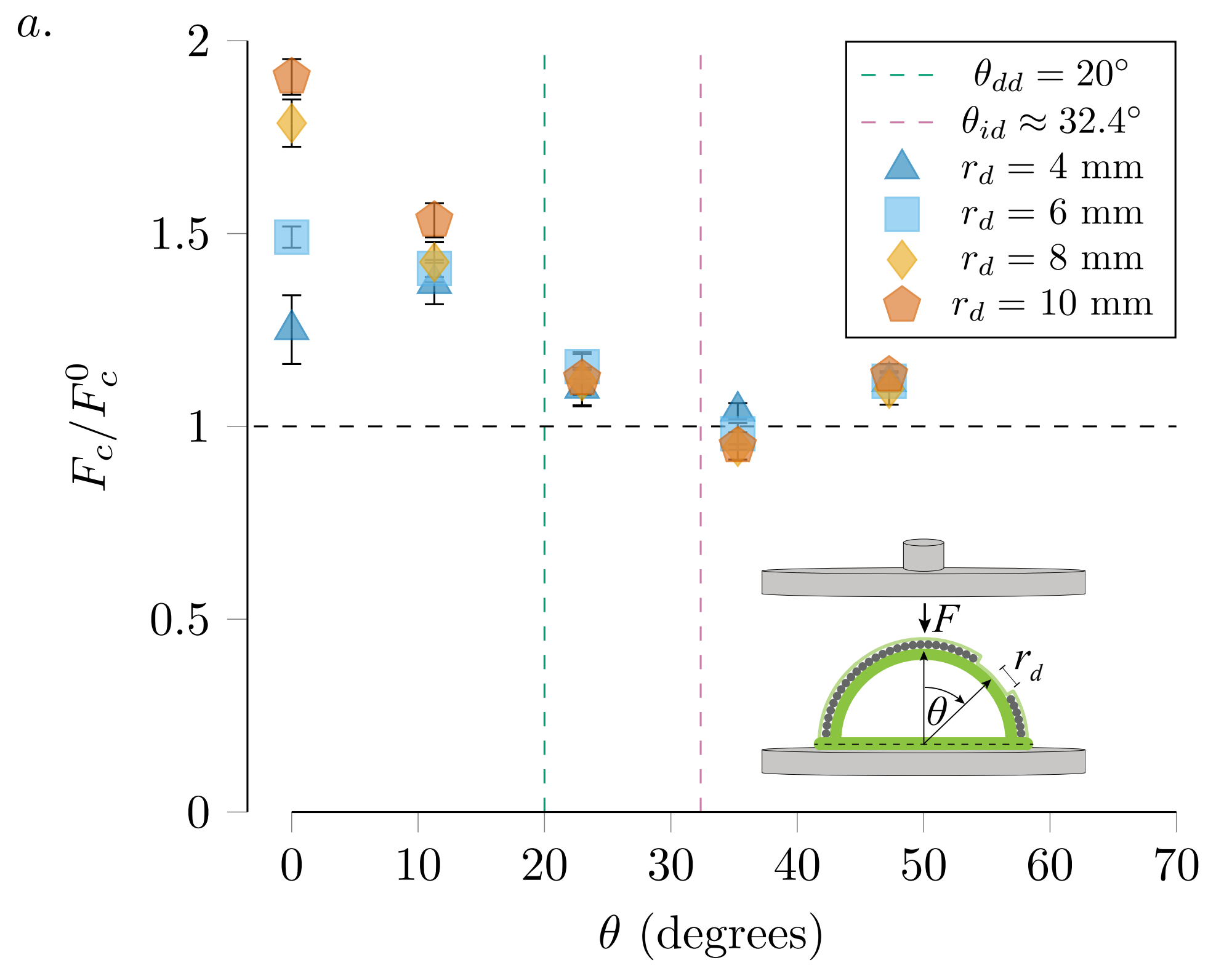
a.
2
1.5
1
0.5
0
F_c/F_c^0
0
10
20
30
40
50
60
70
θ (degrees)
$\theta_{dd} = 20°$
$\theta_{id} \approx 32.4°$
$r_d = 4$ mm
$r_d = 6$ mm
$r_d = 8$ mm
$r_d = 10$ mm
F
r_d
θ